

Healthcare Analytics Dashboard

Final Detailed Analysis Report By Sourav Roychoudhury

Tool: Power BI | Domain: Hospital Operations & Patient Analytics

Executive Summary

This Healthcare Analytics Dashboard was developed using Power BI to analyze hospital operations, patient admissions, financial performance, doctor feedback, bed occupancy, diagnosis trends, and insurance coverage. The dashboard consolidates multiple healthcare KPIs into an interactive report that assists hospital administrators in monitoring operational efficiency, financial health, and patient care.

The dashboard includes interactive slicers, KPI cards, column charts, donut charts, funnel charts, and line charts to provide comprehensive insights into healthcare performance. Overall, the dashboard demonstrates how Business Intelligence can support data-driven decision-making in the healthcare sector.

1. Project Objective

The primary objective of this project is to analyze hospital data and provide meaningful insights regarding:

- Patient admissions and hospital occupancy
- Financial performance and insurance claims
- Doctor performance and patient feedback
- Diagnosis distribution and treatment trends
- Hospital operational efficiency

The dashboard enables healthcare administrators to monitor key performance indicators through an interactive interface.

2. Dataset Overview

The dataset contains patient-level healthcare records. Key variables include:

Variable	Description
Patient ID	Unique patient identifier
Admission / Discharge Date	Hospital stay duration

Follow-up Date	Post-discharge care schedule
Doctor Name	Attending physician
Diagnosis	Medical condition
Bed Occupancy	Bed type and usage
Billing Amount	Total treatment cost
Health Insurance Amount	Insurance coverage
Patient Feedback	Patient satisfaction score
Medical Tests	Tests performed during stay

3. Dashboard Components

KPI Cards

- Admission Date
- Discharge Date
- Follow-up Date
- Total Billing Amount

Interactive Filters

- Patient ID
- Admission Date

Charts

- Bed Occupancy Distribution — Column Chart
- Doctor Performance — Donut Chart
- Diagnosis Distribution — Funnel Chart
- Billing vs Insurance Comparison — Line Chart

4. Data Cleaning & Preparation

Before visualization, the dataset underwent preprocessing to ensure accuracy and consistency. Key steps included:

- Date formatting for admission, discharge, and follow-up dates
- Validation of patient IDs
- Numerical formatting for billing and insurance amounts
- Aggregation of billing values
- Categorization of diagnosis and occupancy data

5. Patient Admission Analysis

The dashboard provides admission-related insights through interactive filters and KPI cards. Hospital administrators can monitor patient inflow, track admission trends, filter individual patient records, and analyze hospital activity over specific periods.

Admission analysis helps hospitals forecast patient demand, allocate medical staff, and improve resource planning.

6. Discharge & Follow-up Analysis

The dashboard tracks discharge dates and follow-up schedules. Monitoring these metrics enables hospitals to reduce readmission rates, improve patient recovery monitoring, and ensure continuity of care.

Recommendation: Automated follow-up reminders can further improve patient satisfaction and reduce missed appointments.

7. Financial Performance Analysis

Total Billing Amount is one of the primary KPIs displayed on the dashboard. This metric measures the total revenue generated from patient treatments.

Monitoring billing performance helps management evaluate hospital profitability, identify revenue trends, plan future investments, and monitor department performance.

8. Health Insurance Analysis

The line chart compares Total Billing Amount and Total Health Insurance Amount across different medical tests. This comparison helps determine the proportion of treatment costs covered by insurance, patient out-of-pocket expenses, and insurance utilization patterns.

Recommendation: Hospitals can negotiate with insurance providers to improve reimbursement rates for frequently performed medical tests.

9. Bed Occupancy Analysis

The column chart displays patient distribution across different bed occupancy categories. Bed occupancy is a critical operational metric.

Occupancy Level	Possible Interpretation
High	Increased patient demand, efficient utilization, potential capacity constraints
Low	Underutilized resources, possible operational inefficiencies

Recommendation: Hospital administrators should monitor occupancy levels regularly to optimize staffing and infrastructure.

10. Doctor Performance Analysis

Doctor performance is evaluated using a donut chart based on cumulative patient feedback scores. Higher feedback scores indicate better patient satisfaction, higher quality of care, and strong physician performance.

Hospital management can use this data to recognize high-performing doctors, identify training opportunities, and improve the overall patient experience.

11. Diagnosis Analysis

The funnel chart visualizes the frequency of different diagnoses across patients. This analysis identifies the most common illnesses, high-demand treatment areas, and disease prevalence within the hospital.

Hospitals can use these insights to plan inventory, allocate specialists, and improve treatment capacity for high-frequency conditions.

12. Billing vs Insurance Comparison

The line chart compares billing amounts with insurance coverage for different medical tests, highlighting:

- Tests with high insurance coverage
- Tests generating higher patient out-of-pocket expenses
- Revenue contribution by service type

Recommendation: Hospitals can evaluate pricing strategies and insurance partnerships based on these findings.

13. Operational Efficiency

The combination of admission, discharge, follow-up, occupancy, and diagnosis metrics enables comprehensive monitoring of hospital operations. Hospital administrators can monitor patient flow, optimize bed utilization, improve discharge planning, and reduce operational bottlenecks.

14. Dashboard Interactivity

The Power BI dashboard supports interactive analysis through the following features:

- Dynamic slicers for Patient ID and Admission Date
- Cross-filtering between visuals
- Drill-down capabilities for detailed analysis
- Real-time KPI updates

These features allow users to perform detailed analysis without modifying the underlying dataset.

Key Business Insights

- The dashboard centralizes operational, financial, and clinical healthcare metrics into a single interactive interface
- Patient admission, discharge, and follow-up tracking improve hospital workflow management
- Bed occupancy analysis supports efficient resource utilization and capacity planning
- Billing and insurance comparisons provide valuable insights into hospital revenue and reimbursement performance
- Doctor feedback analysis helps evaluate physician performance and enhance patient satisfaction
- Diagnosis distribution identifies the most common medical conditions, enabling better inventory and staffing decisions
- Interactive filtering by patient and admission date enables detailed operational analysis

Business Recommendations

1. **Improve Bed Management:** Monitor occupancy trends continuously to optimize bed allocation and reduce waiting times.
2. **Strengthen Financial Monitoring:** Regularly compare billing amounts with insurance reimbursements to identify revenue gaps and improve financial planning.
3. **Enhance Patient Follow-up:** Implement automated reminder systems for follow-up appointments to improve treatment continuity and reduce readmissions.

4. **Optimize Doctor Performance:** Use patient feedback to recognize high-performing physicians and identify opportunities for additional training or support.
5. **Analyze Diagnosis Trends:** Allocate medical staff, equipment, and inventory based on the most frequently occurring diagnoses.
6. **Improve Insurance Partnerships:** Evaluate reimbursement rates for commonly performed medical tests and negotiate better insurance agreements where necessary.
7. **Leverage Interactive Reporting:** Encourage hospital administrators and department heads to use the dashboard for regular performance reviews, enabling faster and more informed decision-making.

Conclusion

The Healthcare Analytics Dashboard effectively demonstrates how Power BI can transform raw hospital data into actionable business intelligence. By integrating patient records, financial metrics, operational KPIs, and clinical insights into a single interactive dashboard, the project supports informed decision-making across multiple aspects of healthcare management.

The analysis highlights opportunities to improve patient care, optimize hospital resources, strengthen financial performance, and enhance operational efficiency. This project showcases practical skills in data visualization, dashboard design, KPI development, and healthcare analytics, making it a valuable addition to a data analytics portfolio.